

Quantum Mechanics II Exercice 2

Idan Stark

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1 Analytical problems

1.1 Ising model with nearest + next nearest interaction

The hamiltonian for the Ising model with nearest + next nearest interaction is

$$H = -J_1 \sum_i \sigma_i \sigma_{i+1} - J_2 \sum_i \sigma_i \sigma_{i+2} - B \sum_i \sigma_i \quad (1)$$

Since the boundaries are negligible in the thermodynamic limit, we can select periodic boundary conditions $N+1 = 1, N+2 = 2$. The partition function is then

$$\begin{aligned} Z_N &= \sum_{\sigma_1=\pm 1} \cdots \sum_{\sigma_N=\pm 1} \exp \left[\beta J_1 \sum_i \sigma_i \sigma_{i+1} + \beta J_2 \sum_i \sigma_i \sigma_{i+2} + \beta B \sum_i \sigma_i \right] = \\ &= \sum_{\sigma_1=\pm 1} \cdots \sum_{\sigma_N=\pm 1} \prod_i \exp (\beta J_1 \sigma_i \sigma_{i+1} + \beta J_2 \sigma_i \sigma_{i+2} + \beta B \sigma_i) \end{aligned} \quad (2)$$

To create a transfer matrix, let us define a superspin $|\bar{\sigma}_i\rangle \in \mathbb{R}^4$ out of every two consecutive spins:

$$|\uparrow\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad |\uparrow\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad |\downarrow\uparrow\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad |\downarrow\downarrow\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \quad (3)$$

Which means the exponent in equation 2 is

$$\exp (\beta J_1 \sigma_i \sigma_{i+1} + \beta J_2 \sigma_i \sigma_{i+2} + \beta B \sigma_i) = \langle \bar{\sigma}_i | M | \bar{\sigma}_{i+1} \rangle \quad (4)$$

Where M is the transfer matrix:

$$M = \exp \left[\beta \begin{pmatrix} J_1 + J_2 + B & J_1 - J_2 + B & & \\ -J_1 + J_2 + B & -J_1 - J_2 + B & & \\ & & -J_1 - J_2 - B & -J_1 + J_2 - B \\ & & J_1 - J_2 - B & J_1 + J_2 - B \end{pmatrix} \right] \quad (5)$$

Naming the four vectors from equation 3 $b_1, \dots, b_4$, the partition function is

$$Z_N = \sum_{\bar{\sigma}_1=b_1 \dots b_4} \cdots \sum_{\bar{\sigma}_N=b_1 \dots b_4} \langle \bar{\sigma}_1 | M | \bar{\sigma}_2 \rangle \langle \bar{\sigma}_2 | M | \bar{\sigma}_3 \rangle \cdots \langle \bar{\sigma}_N | M | \bar{\sigma}_1 \rangle = \text{Tr}(M^{N/4}) \quad (6)$$

Where in the last equation we used the fact that M is translationally invariant. Since the trace is base independent, we can express it as the sum of the matrix's eigenvalues, and in the thermodynamic limit, all eigenvalues except the largest one will be negligible, and the partition function is

$$\lim_{N \rightarrow \infty} Z_N = \lambda_{\max}^N \quad (7)$$

This result makes sense because it gives us extensive free energy.

1.2 Ising model as a Lattice Gas

Given seven sites and three particles that could be in those sites, there is a $\binom{7}{3}$ options for their configuration, but we can make the situation much simpler. The first thing to notice is the translation invariance, so we can assume the first particle is in the first position, and simply multiply the options by 7 at the end. This should leave us with $\binom{7}{3}/7 = 5$ options, which is a very manageable number. Now let's look at the positions of the other two particles. We don't want to overcount by saying that placing the particles in positions (1, 6, 7) is distinct from placing them in (1, 2, 3). Those two situations are the same under translations, which we already accounted for. We can find duplicates by enumeration and remove them. The positions that are left after removing all duplicates are

$$b_1 = (1, 2, 3) \quad b_2 = (1, 2, 4) \quad b_3 = (1, 2, 5) \quad b_4 = (1, 2, 6) \quad b_5 = (1, 3, 5) \quad (8)$$

Using the Ising hamiltonian for each configuration, the magnetic field term gives $-3B$ for every configuration, while the interaction term is positive if the two sites are both occupied or both unoccupied. i.e. we start from $-7J$ and gain $2J$ for every wall:

$$Z_1 = e^{3\beta J + 3\beta B} \quad Z_2 = Z_3 = Z_4 = e^{-\beta J + 3\beta B} \quad Z_5 = e^{-5\beta J + 3\beta B} \quad (9)$$

In other words we have one situations with two walls, three situations with four walls and one situation with six walls. Summing over each of them and multiplying by seven for the translation invariance, we get:

$$\begin{aligned} Z &= e^{3\beta J + 3\beta B} + 3e^{-\beta J + 3\beta B} + e^{-5\beta J + 3\beta B} = \\ &= e^{3\beta B} (e^{3\beta J} + 3e^{-\beta J} + e^{-5\beta J}) \end{aligned} \quad (10)$$

2 Numeric Problems

2.1 Easy Numeric

2.1.1 The free energy and the partition function